vertices in the graph. On the contrary, several “restarts”, from different starting points are in general needed to visit all the vertices in a graph: we’ll show this while discussing DFS.

Initially, however, we will focus on a specific use case: considering the starting point as “given” (externally chosen, without the algorithm knowing much about it) and traversing the graph from it. Based on these assumptions, we will discuss the two most common traversal strategies used on graphs.

14.3.1 Optimizing Delivery Routes

It’s time to go back to the problem with which we have introduced this chapter: we have a single delivery to perform from a source point, the warehouse or factory where goods are stocked, to a single destination, the customer’s address.

The hypothesis that we handle deliveries one by one is, obviously, already a simplification: in the general case, it would be too expensive, and delivery companies try to gather together orders from the same warehouse to close-by destinations, to spread the costs (gas, employee’s time, etc…) over several orders.

Finding the best route passing through several points, however, is a computationally-hard problem[4]; the best route for the single-source, single-destination case, conversely, can be found efficiently.